# Gate Alignment Before Causal Branch Modularity

## Branch Specialization Phase 3 Toy Checkpoint

Branch specialization research project

2026-05-22

Checkpoint reason: trajectory evidence identifies when role-informative routing pressure becomes causal branch modularity.

# Current Question

The annealed-label experiment showed that labels through step 400 fail, labels through step 800 partially persist, and labels through step 1200 persist more strongly.

**Question.** What changes between step 400 and step 800?

Two possibilities:

- the router gate itself may not be role-aligned until late;
- the gate may align early, while branch computations need more time to become causally separable.

# Setup

- Task: `bidirectional_lookup`
- Branches: two 64-dim attention branches
- Seeds: 5
- Training: 1600 optimizer updates
- Trajectory steps: 0, 50, 100, 200, 400, 401, 600, 800, 801, 1000, 1200, 1201, 1400, 1600

Conditions:

| Condition | Role-label schedule |
|---|---|
| unlabeled | entropy 0.05 plus balance 1.0, no labels |
| end 400 | weak labels for steps $< 400$ |
| end 800 | weak labels for steps $< 800$ |
| end 1200 | weak labels for steps $< 1200$ |
| always | weak labels for the full run |

# Metric Glossary

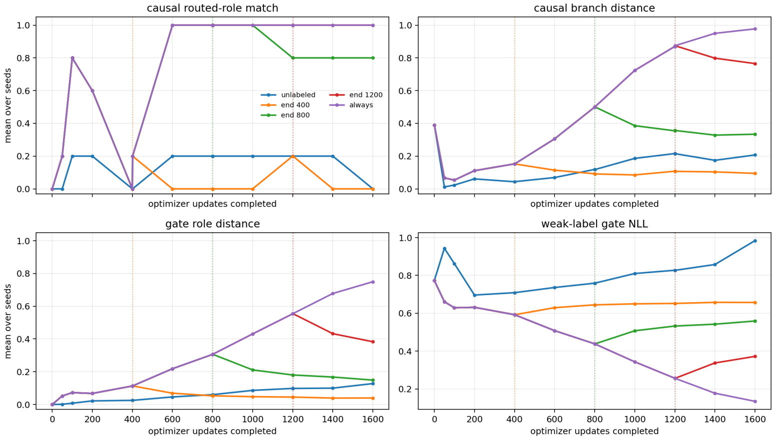
**Gate role distance** Distance between local-role and induction-role router probability distributions. This measures where the gate sends tokens.

**Gate routed match** Fraction of seeds where the gate maps local to branch 0 and induction to branch 1.

**Causal branch distance** Distance between branch-use distributions inferred by branch ablation. This measures where the computation lives.

**Routed causal match** Fraction of seeds where causal ablation maps local to branch 0 and induction to branch 1.

# Trajectory Plot




Dotted vertical lines mark label-removal steps. The key transition is between step 400 and step 600: gates are already aligned, but causal branch modularity appears only under continued labels.

# Step 400 Is Not Enough

| Schedule / step | Gate match | Gate dist. | Causal match | Branch dist. | Ind. acc. |
|---|---|---|---|---|---|
| end 400 / 400 | 1.00 | 0.1127 | 0.00 | 0.1525 | 0.9990 |
| end 400 / 600 | 1.00 | 0.0688 | 0.00 | 0.1141 | 0.9994 |
| end 400 / 1600 | 0.60 | 0.0395 | 0.00 | 0.0945 | 0.9992 |

By step 400, task performance and gate alignment are already high. Removing labels there prevents causal branch modularity from consolidating.

# Continued Labels Make The Causal Split

| Schedule / step | Gate dist. | Causal match | Branch dist. | Gate NLL |
|---|---|---|---|---|
| end 800 / 400 | 0.1127 | 0.00 | 0.1525 | 0.5927 |
| end 800 / 600 | 0.2183 | 1.00 | 0.3056 | 0.5085 |
| end 800 / 800 | 0.3058 | 1.00 | 0.4996 | 0.4388 |
| end 800 / 1600 | 0.1488 | 0.80 | 0.3337 | 0.5593 |

The causal split appears by step 600 if labels continue. After label removal at step 800, the split mostly persists but weakens.

# Late Removal Preserves A Stronger Split

| Schedule / step | Gate dist. | Causal match | Branch dist. | Gate NLL |
|---|---|---|---|---|
| end 1200 / 1200 | 0.5539 | 1.00 | 0.8701 | 0.2571 |
| end 1200 / 1600 | 0.3831 | 1.00 | 0.7652 | 0.3725 |
| always / 1200 | 0.5539 | 1.00 | 0.8701 | 0.2571 |
| always / 1600 | 0.7497 | 1.00 | 0.9773 | 0.1351 |

Removing labels after a strong causal split preserves top-branch assignment, but continuous labels keep strengthening separation.

# Interpretation

**Result**

- Gate alignment is not sufficient evidence of functional modularity.
- The gate becomes role-aligned before the branches become causally separable.
- Branch modularity consolidates later under sustained role-aligned routing pressure.

**Caveat**

- Early routed-match values before the task is solved should not be interpreted strongly. The decision-relevant comparison is step 400 onward, when both roles have near-perfect accuracy.

# Project-Level Update

The toy evidence now supports a causal-consolidation framing:

*Structural routing plus role-informative training pressure can produce functional branch modularity, but the branch computation lags behind gate alignment.*

The paper-facing question becomes:

*Under what training signals and time windows does structural modularity become causal functional modularity?*

# Decision And Next Step

**Decision.** The current toy branch is coherent enough. Stop broad toy sweeps of router regularizers and anneal times.

**Next step.** Move to a less hand-designed routed attention setting or a small real transformer task:

- keep role probes separate from causal branch-ablation or patching;
- test whether gate alignment again precedes causal modularity;
- compare structural routing, weak role labels, and unlabeled pressure.

# Provenance

## Source files

- scripts/toy_branch_isolation_intervention.py
- scripts/analyze_router_trajectory_experiment.py
- doc/phase3_toy_router_trajectory.md

## Copied data in this deck folder

- data/trajectory_by_step.csv
- data/trajectory_rows.csv
- figures/router_trajectory_metrics.png

## Rebuild

- python3 -u scripts/analyze_router_trajectory_experiment.py
- python3 .../compile_latex_deck.py presentations/2026-05-22-1745-router-trajectory/router_trajectory_checkpoint.tex